SOFTWARE REQUIREMENTS SPECIFICATION

VERSION 2.0

REAL-TIME ATTENDANCE MONITERING SYSTEM (RAMS)

Prepared By

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document provides a comprehensive and structured description of the Digital Attendance Management System. The purpose of this document is to define the functional and non-functional requirements, system constraints, and operational behaviour of the system in a precise and unambiguous manner. It serves as a formal agreement between stakeholders, developers, and evaluators, ensuring that all parties share a common understanding of the system's objectives and expected outcomes.

The document also acts as a reference point throughout the software development lifecycle, guiding design decisions, implementation strategies, and validation processes.

1.2 Scope

The Digital Attendance Management System is a web-based software application designed to streamline and automate the process of attendance tracking in educational institutions. The system enables faculty members to efficiently mark attendance, allows students to monitor their academic attendance status in real time, and provides administrative control for managing institutional data and policies.

The system covers core functionalities including user authentication, subject-wise attendance management, performance tracking, eligibility evaluation based on institutional rules, and centralized data management. It is intended to reduce manual effort, improve data accuracy, and provide transparency across all levels of users.

1.3 Acronyms and Abbreviations

Term / Acronym	Definition / Full Form
Attendance	The record of a student's presence or absence for a particular class session.
Attendance Percentage	The ratio of classes attended by a student to the total number of classes conducted for a subject, expressed as a percentage.
Attendance Marks	Marks awarded to a student based on attendance percentage as per institutional rules.
Attendance Rules	Predefined institutional policies that specify attendance thresholds for examination eligibility and mark allocation.
Administrator	A system user responsible for managing student records, subjects, faculty assignments, and attendance rules.

Term / Acronym	***Definition / Full Form***
Class Session	*A single occurrence of a class conducted for a subject on a specific date and time.*
Examination Eligibility	*The qualification status of a student to appear in examinations based on attendance criteria.*
Faculty	*A system user responsible for conducting classes and recording attendance for assigned subjects.*
Functional Requirement (FR)	*A specification that describes a function or behavior that the system must perform.*
Non-Functional Requirement (NFR)	*A specification that defines system qualities such as security, availability, and performance.*
Student	*A system user enrolled in one or more subjects whose attendance is monitored by the system.*
Subject	*An academic course or unit offered by the institution for which attendance is recorded.*
User	*Any individual who interacts with the system, including students, faculty members, and administrators.*
Web-Based Application	*A software system accessed through a web browser over a network.*
SRS	*Software Requirements Specification*
UI	*User Interface*
DB	*Database*
API	*Application Programming Interface*
CRUD	*Create, Read, Update, Delete*
SQL	*Structured Query Language*

1.4 References

- *IEEE Std 830–1984 – Software Requirements Specification Guidelines*
- *Academic Attendance Policies (University of Delhi)*

1.5 Overview

This document is organized into multiple sections covering system overview, user interaction, functional requirements, system design, database schema, and traceability. Each section builds upon the previous one to provide a detailed understanding of the system.

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2. Overall Description

2.1 Product Perspective

The Digital Attendance Management System is designed as a standalone web-based application operating within a client-server architecture. The frontend component provides an interactive user interface accessible through web browsers, while the backend handles business logic, data processing, and database interactions.

The system integrates multiple modules including authentication, attendance tracking, reporting, and administrative control. It maintains a centralized database that stores all user and attendance-related data. While the system currently operates independently, it is designed with extensibility in mind, allowing future integration with institutional ERP systems or learning management platforms.

2.2 Product Functions

The system provides a comprehensive set of functionalities aimed at efficient attendance management and academic monitoring. These functions include, but are not limited to:

- *Secure user authentication for students, faculty, and administrators*
- *Storage and management of student and faculty information*
- *Subject-wise attendance tracking and real-time updates*
- *Automated calculation of attendance percentage*
- *Determination of eligibility for examinations based on predefined thresholds*
- *Calculation of required classes to meet attendance criteria (67%, 75%, 85%)*
- *Faculty interface for marking attendance in a structured tabular format*
- *Administrative control for managing user data, permissions, and system rules*

These functions collectively ensure that the system not only records attendance but also provides actionable insights to users.

2.3 User Characteristics

The system is intended for three primary categories of users:

Students

Students are expected to have basic familiarity with web applications. Their interaction with the system is primarily read-only, focusing on viewing attendance statistics, eligibility status, and academic performance.

Faculty

Faculty members possess moderate technical proficiency and are responsible for marking attendance, managing subject-level data, and reviewing student records.

Administrator

Administrators have advanced access privileges and are responsible for maintaining system data, configuring rules, and managing permissions. They ensure the integrity and correctness of system operations.

2.4 Operating Environment

The system operates in a web-based environment and requires:

- *A modern web browser (Chrome, Edge, Firefox)*
- *Backend server (Node.js / Django / similar)*
- *Relational database system (MySQL / PostgreSQL)*

2.5 Constraints

The system is subject to several constraints:

- *Requires continuous internet connectivity*
- *Dependent on accurate data entry by faculty*
- *Must adhere to institutional attendance policies*
- *Limited by browser compatibility and server performance*

2.6 Assumptions and Dependencies

The system assumes that:

- *Student and faculty data are pre-registered*
- *Subjects are correctly mapped to both students and faculty*
- *Attendance policies (67%, 75%, 85%) are predefined*
- *Semester duration and total classes are known*

2.7 User Use Cases

This section describes the interaction between users and the system.

2.7.1 Use Case: Student Dashboard and Attendance Analysis

Field	***Details***
Description	*This use case allows students to view their academic attendance details in a subject-wise manner.*
Actors	*Student*
Preconditions	*Student must be authenticated*
Main Flow	*1.Student logs into the system* *2. System displays dashboard with enrolled subjects* *3. Student selects a subject* *4. System retrieves attendance data* *5. System calculates:* *• Attendance percentage* *• Eligibility status* *• Required classes for thresholds* *6. System displays results*
Postconditions	*Student receives updated attendance insights*

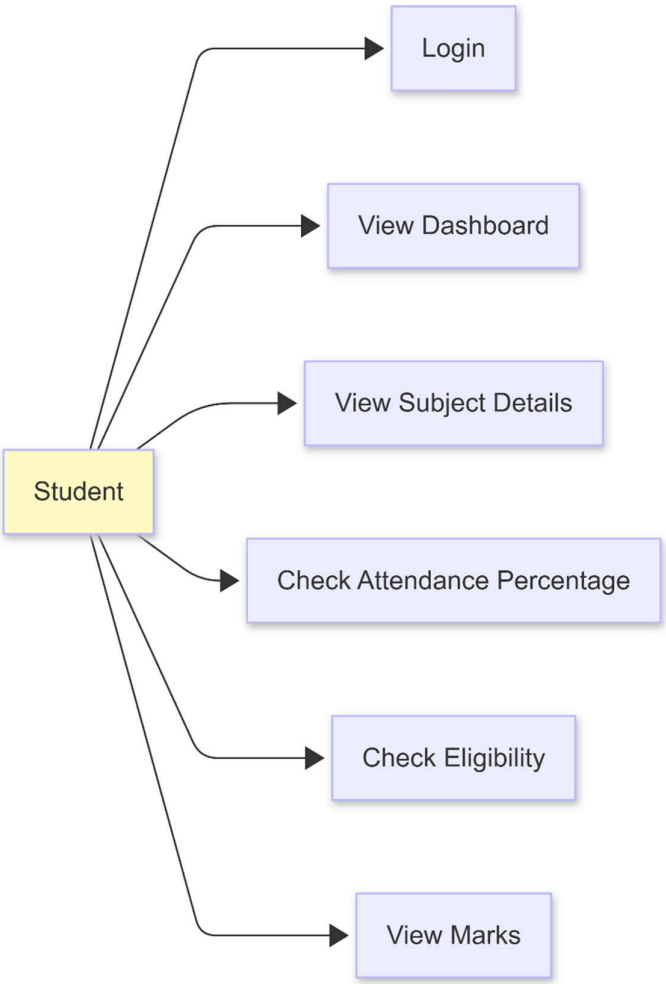

Fig 1.1: Student use case diagram

2.7.2 Use Case: Faculty Attendance Management

Field	***Details***
Description	*This use case enables faculty members to mark and manage attendance for students.*
Actors	*Faculty*
Preconditions	*Faculty must be authenticated*
Main Flow	1. *Faculty logs in* 2. *System displays assigned subjects* 3. *Faculty selects a subject* 4. *System loads structured attendance sheet* 5. *Sheet displays:*

Field	Details
	- Student name - Roll number - Previous attendance - Current percentage 6. Faculty marks attendance 7. System validates and stores data 8. System updates attendance statistics
Postconditions	Attendance is recorded and updated

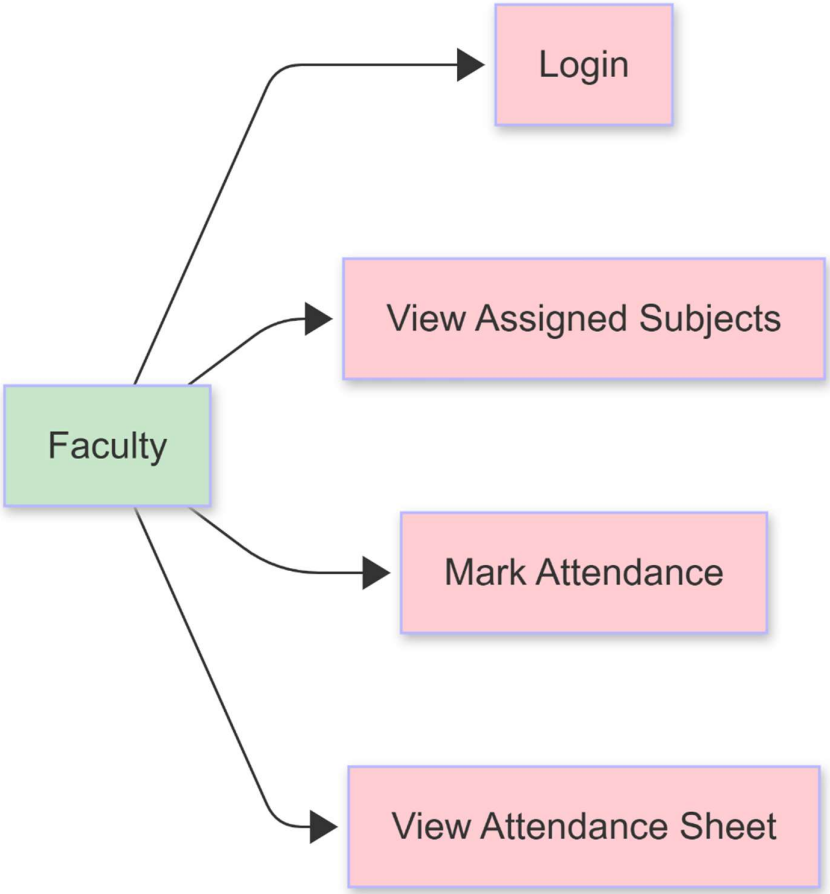

Fig 1.2: Faculty use case diagram

2.7.3 Use Case: Administrator Control and Configuration

Field	Details
Description	Allows administrators to manage system data and rules.

Field	Details
Actors	Administrator
Preconditions	Administrator must be authenticated
Main Flow	1. Admin logs in 2. Admin selects operation (Manage students, manage faculty, configure rules, Assign permissions); 3. System processes updates 4. Database is updated
Postconditions	System data and rules are updated

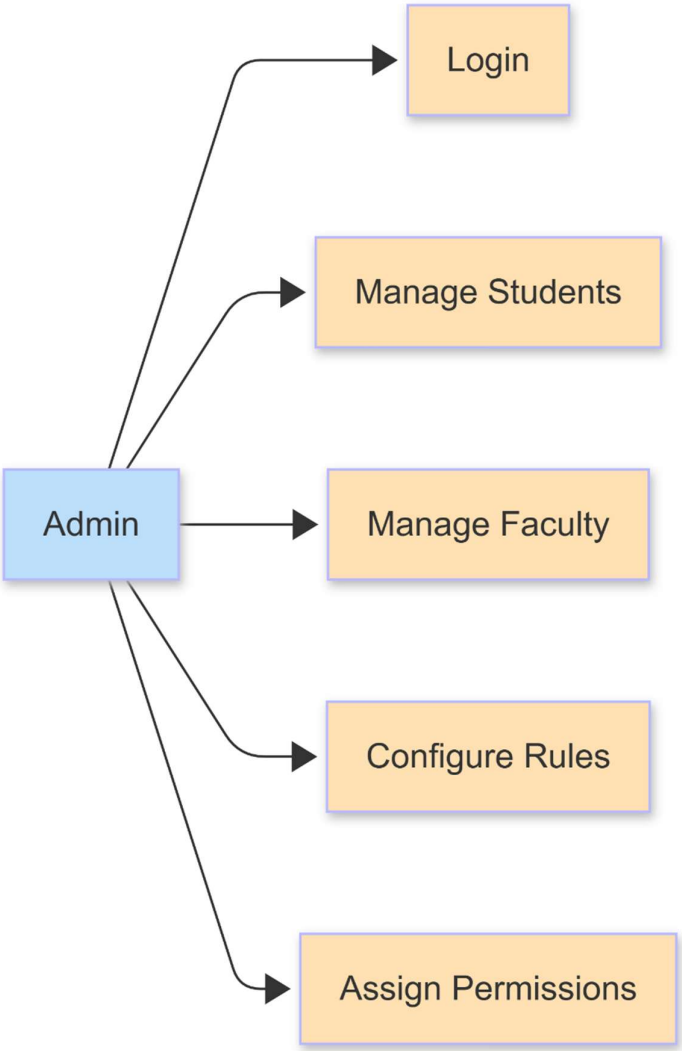

Fig 1.3: Admin use case diagram

3. Specific Requirements

3.1 Functional Requirements

3.1.1: Authentication System

The system shall provide secure login functionality for all users. Credentials shall be validated before granting access.

3.1.2: Student Data Management

The system shall store detailed student information including academic and personal identifiers.

3.1.3: Faculty Data Management

The system shall maintain faculty records and subject assignments.

3.1.4: Attendance Recording

The system shall allow faculty to mark attendance through a structured interface. Attendance data shall be validated, stored, and immediately reflected in student records.

3.1.5: Attendance Calculation

The system shall compute attendance percentage dynamically using stored attendance records.

3.1.6: Eligibility and Requirement Calculation

The system shall determine eligibility based on predefined thresholds and calculate required attendance for future compliance.

3.1.7: Dashboard Display

The system shall present attendance, eligibility, and performance data in a clear and structured format.

3.1.8: Administrative Controls

The system shall allow administrators to manage data, permissions, and system rules.

3.2 Performance Requirements

The system shall ensure fast response times, with dashboard loading and attendance updates occurring within acceptable time limits under normal load conditions.

3.3 Design Constraints

The system shall use a relational database with normalized structure to ensure data integrity and efficient querying.

3.4 External Interface Requirements

User Interface

The interface shall be simple, structured, and intuitive, resembling spreadsheet-like layouts for attendance marking.

Software Interfaces

The system shall interact with backend APIs and database systems.

3.5 Non-Functional Requirements

- *Usability: Easy to learn and operate*
- *Reliability: Accurate data processing*
- *Security: Role-based access control*
- *Scalability: Support multiple users simultaneously*

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4. Software Design

4.1 System Architecture

The Digital Attendance Management System follows a modular client–server architecture designed to ensure scalability, maintainability, and efficient data handling within a college environment.

The system is divided into three primary architectural layers: the Presentation Layer, the Application Layer, and the Data Layer.

Presentation Layer (Frontend)

This layer represents the user interface of the system, accessible through web browsers. It provides separate dashboards for students, faculty, and administrators. The interface is designed to be simple and structured, allowing users to perform operations such as viewing attendance, marking attendance, and managing records with minimal complexity.

Application Layer (Backend Server)

The backend layer is responsible for handling all business logic and system operations. It processes user requests, performs validation, computes attendance percentages, determines eligibility, and manages communication between the frontend and the database.

This layer also ensures that operations such as attendance marking, data updates, and rule enforcement are executed reliably and consistently.

Data Layer (Database System)

The data layer consists of a relational database that stores all system data, including student records, faculty details, subject mappings, attendance logs, and marks. It ensures data integrity through structured relationships and constraints.

The interaction between these layers occurs through secure communication protocols, ensuring that user actions are processed efficiently and data is retrieved or updated accurately.

This layered architecture promotes separation of concerns, allowing independent modification and scalability of system components without affecting overall functionality.

4.2 Detailed Design

The detailed design of the system is based on object-oriented principles, ensuring modularity, reusability, and maintainability.

The system is composed of several core entities, each representing a real-world component of the attendance management process.

Core Classes / Entities

The major entities in the system include:

- ***Student**: Represents a student with attributes such as name, enrollment number, roll number, academic details, and associated subjects.*
- ***Faculty**: Represents faculty members responsible for teaching subjects and marking attendance.*
- ***Subject**: Represents academic subjects linked to both students and faculty.*
- ***Attendance**: Represents attendance records for each student per subject per date.*
- ***Marks**: Stores academic performance data for each student in specific subjects.*
- ***Rules**: Defines attendance thresholds and eligibility criteria.*

Each entity encapsulates its attributes and interacts with other entities through defined relationships, ensuring proper data flow and consistency.

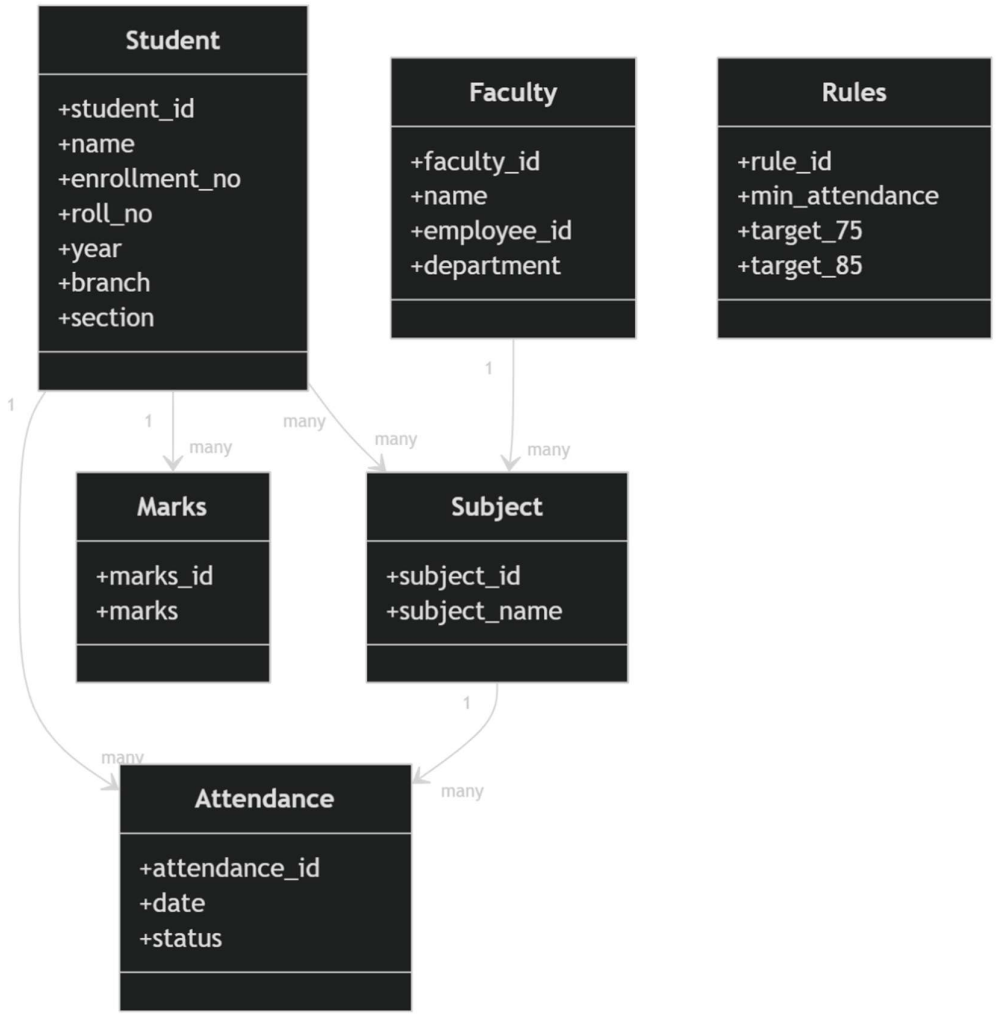

Fig 2: Class Diagram

4.3 Sequence Diagrams

4.3.1 Sequence Flow – Student Viewing Attendance

1. Student logs into system
2. Request sent to backend
3. Backend fetches student subjects
4. Student selects subject
5. Backend retrieves attendance records
6. System calculates percentage and eligibility
7. Data returned and displayed

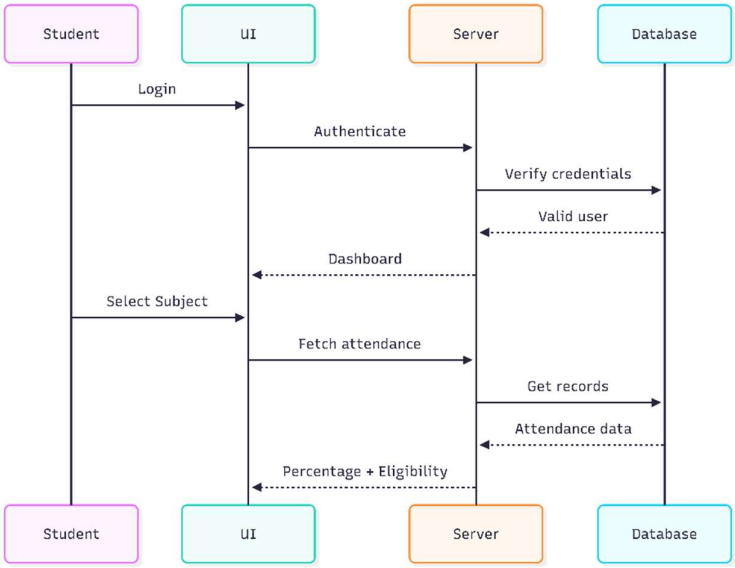

Fig 3.1: Sequence Diagram – Students

4.3.2 Sequence Flow – Faculty Marking Attendance

1. Faculty logs in
2. Selects subject
3. Backend retrieves student list
4. Attendance sheet displayed
5. Faculty marks attendance
6. Data sent to backend
7. Database updated

8. *Attendance recalculated*

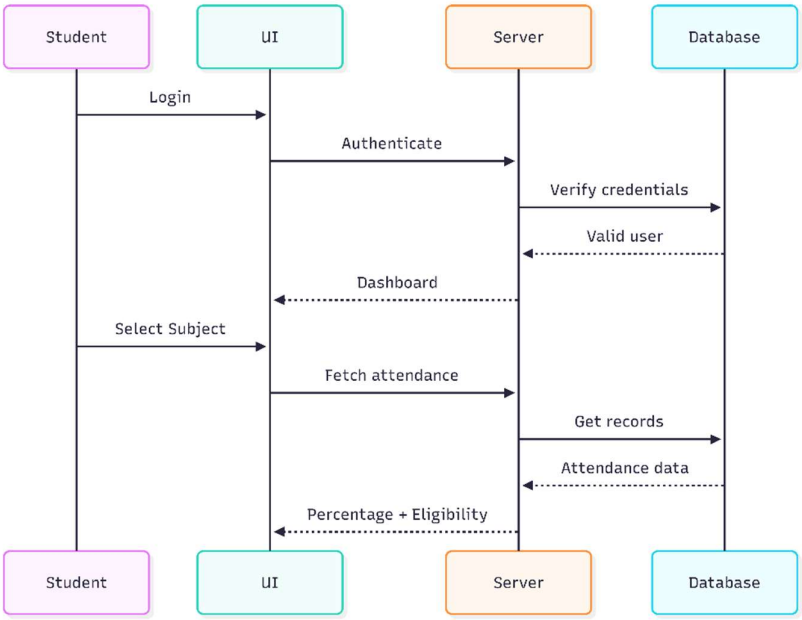

Fig 3.2: Sequence Diagram - Faculty

4.3.3 Sequence Flow – Admin Updating Data

1. *Admin logs in*
2. *Selects data management*
3. *Inputs updated records*
4. *Backend validates data*
5. *Database updated*
6. *Confirmation returned*

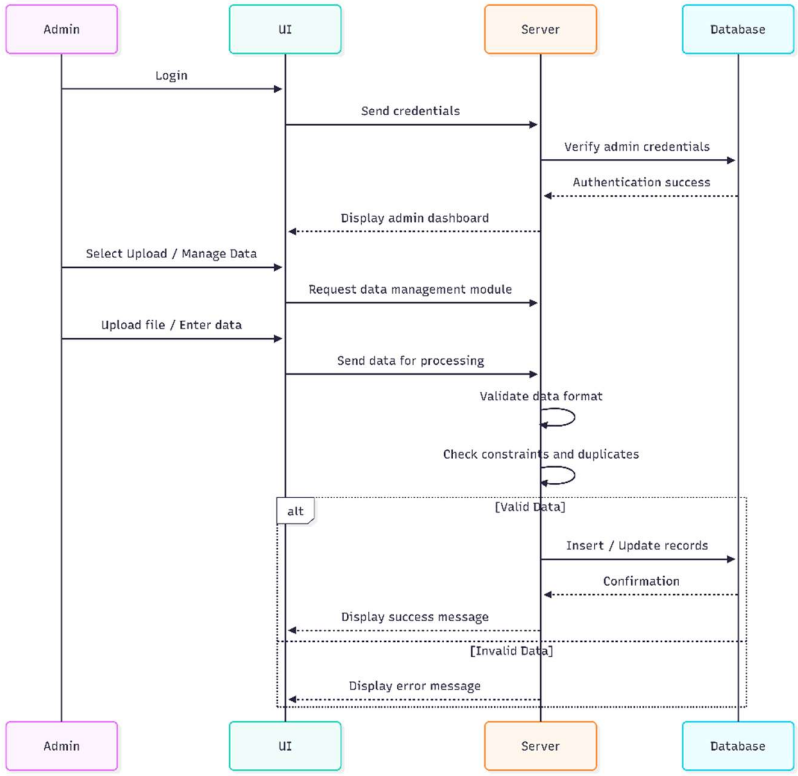

Fig 3.3: Sequence Diagram – Admin

4.4 Database Schema

The database is designed using a relational model with normalized entities:

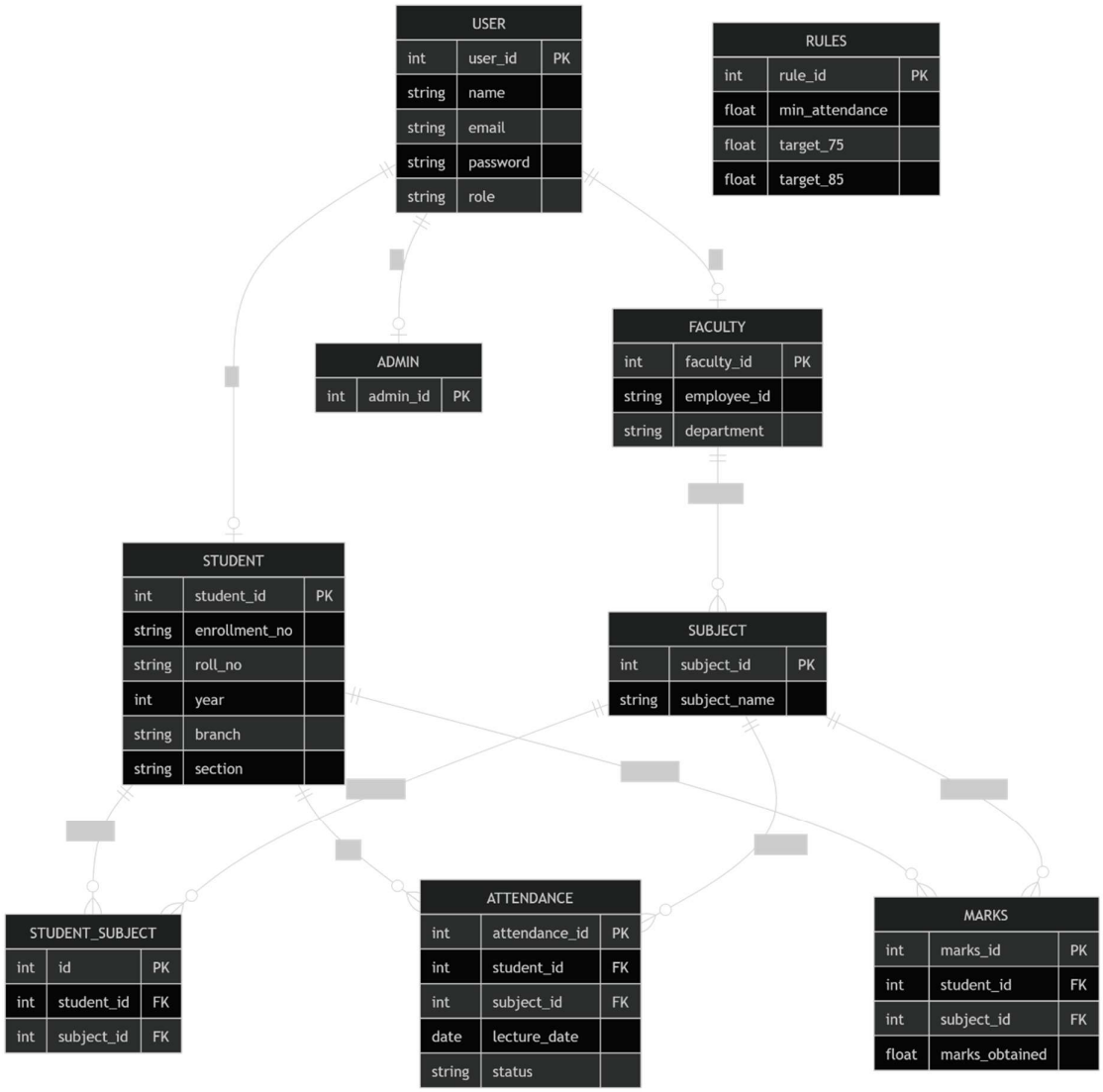

Fig 4: EER Diagram

Students

- *student_id (PK)*
- *name*
- *enrollment_no*
- *roll_no*
- *year*
- *branch*
- *section*

Faculty

- *faculty_id (PK)*
- *name*
- *employee_id*
- *department*

Subjects

- *subject_id (PK)*
- *subject_name*
- *faculty_id (FK)*

Student_Subject

- *id (PK)*
- *student_id (FK)*
- *subject_id (FK)*

Attendance

- *attendance_id (PK)*
- *student_id (FK)*
- *subject_id (FK)*
- *date*
- *status*

Marks

- *marks_id (PK)*
- *student_id (FK)*
- *subject_id (FK)*
- *marks*

4.5 Security Design

The Digital Attendance Management System incorporates multiple security mechanisms to ensure data confidentiality, integrity, and controlled access.

4.5.1 Authentication and Authorization

The system implements secure user authentication using credential-based login mechanisms. Each user (student, faculty, administrator) is assigned a role, and access to system features is controlled through role-based access control (RBAC).

- *Students: Read-only access to personal data*
- *Faculty: Attendance marking and subject management*
- *Admin: Full control over system data and permissions*

Session management techniques are used to maintain secure user sessions and prevent unauthorized access.

4.5.2 Data Security Measures

- *Sensitive data such as passwords are stored in encrypted form*
- *Communication between client and server is secured using HTTPS*
- *Input validation is performed to prevent invalid or malicious data*

4.5.3 Protection Against Common Attacks

The system is designed to protect against common vulnerabilities:

- ***SQL Injection**: Prevented using parameterized queries*
- ***Cross-Site Scripting (XSS)**: Prevented through input sanitization*
- ***Unauthorized Access**: Prevented through strict role validation*

4.5.4 Data Integrity

The database enforces referential integrity through primary and foreign key constraints. This ensures consistency in relationships between students, subjects, and attendance records.

5.0 Traceability

Requirement ID	***Requirement Description***	***Design ID***	***Design Artifact***
FR-1	*User authentication for all users*	*D-1*	*Authentication Module, Sequence Diagram (Login)*

Requirement ID	Requirement Description	Design ID	Design Artifact
FR-2	Student data storage and management	D-2	Database Schema (Student Entity), Admin Module
FR-3	Faculty data storage and management	D-3	Database Schema (Faculty Entity), Admin Module
FR-4	Attendance marking by faculty	D-4	Attendance Module, Sequence Diagram (Faculty Attendance)
FR-5	Attendance percentage calculation	D-5	Calculation Logic, Backend Processing
FR-6	Eligibility determination based on thresholds	D-6	Rules Engine, Calculation Module
FR-7	Student dashboard display	D-7	UI Dashboard, Sequence Diagram (Student View)
FR-8	Admin data upload and management	D-8	Admin Module, Sequence Diagram (Admin Upload)
FR-9	Validation of uploaded data	D-9	Validation Module
FR-10	Storage of attendance per lecture	D-10	Database Schema (Attendance Entity)
FR-11	Storage of student marks	D-11	Database Schema (Marks Entity)
FR-12	Subject allocation to students and faculty	D-12	Database Schema (Subject & Mapping Tables)
FR-13	Role-based access control	D-13	Security Module
FR-14	Configuration of attendance rules	D-14	Rules Module
FR-15	Viewing attendance history	D-15	UI Module, Database Queries

6.0 Future Enhancements

The Digital Attendance Management System, in its current form, provides a comprehensive solution for managing student attendance, academic tracking, and administrative control within a college environment. However, there are several areas where the system can be further enhanced to improve functionality, usability, and scalability in future versions.

One potential enhancement involves the integration of automated attendance marking mechanisms. Instead of relying solely on manual input by faculty members, the system can be extended to support technologies such as QR code-based attendance or biometric verification systems. This would significantly reduce manual effort and improve the accuracy and authenticity of attendance records.

Another important enhancement is the introduction of advanced analytics and reporting features. The system can be expanded to generate detailed reports on student attendance trends, subject-wise performance, and overall academic progress. These analytics can assist faculty and administrators in identifying patterns, such as consistently low attendance in specific subjects or at-risk students, enabling timely intervention.The system can also be enhanced by incorporating a notification and alert mechanism. Students can receive automated alerts regarding low attendance percentages, approaching eligibility thresholds, or upcoming academic deadlines. Similarly, faculty and administrators can be notified about irregular attendance patterns or data inconsistencies, ensuring proactive system usage.

In terms of user experience, the system can be extended to include a mobile application interface. A dedicated mobile platform would provide greater accessibility and convenience, allowing users to access attendance data and perform necessary operations on handheld devices. This would be particularly beneficial for students who frequently rely on mobile devices for academic updates.

Another area of improvement is the enhancement of data import and export capabilities. The current system can be extended to support bulk data handling through structured file formats such as spreadsheets, enabling administrators to efficiently manage large datasets. Additionally, exporting attendance and performance reports in formats such as PDF or CSV can improve usability for record-keeping and reporting purposes.

From a security perspective, future versions of the system can incorporate advanced authentication mechanisms such as multi-factor authentication. This would enhance system security and ensure that sensitive academic data is protected against unauthorized access.

Finally, the system can be designed to support integration with other academic tools and platforms used within the institution, such as timetable systems or internal portals. While the current system operates independently, such integrations can improve overall workflow efficiency and data consistency across platforms.

These enhancements, when implemented, would significantly improve the system's capability, making it more robust, user-friendly, and adaptable to evolving academic requirements.